

Team turn — 2026-07-06 (converge-to-bank). Earlier issues resolved by the arena redirect; three live count-movers. Push by DERIVING forward — a banked-then-retracted count is worse than an honest 8.

date + SOD. Count 8 observables + 2 structural banks (Peirce T2511, descent closure).

Status: how we're doing (honest)

The arena redirect resolved the arc's tangle. Earlier issues, now closed: - curvature-vs-Laplacian flip → resolved: the down-ladder is neither — it's the $SO(4,2)$ conformal Casimir $\Delta(\Delta-4)$ (object pinned). - genus-flip swamp → resolved: p cancels in ratios; book-dependency gone (linear algebra). - one-operator conflation → resolved: masses = conformal Casimir ($SO(4,2)$); mixing = curvature (D_{IV}^5), holographic dual. Cleanly split. - mixing arena → resolved to a decidable test ($SO(4,2)$ OPE = $5/2$).

The down-ladder is the strongest it has ever been — and the auditor's measure of how strong: Grace applied one map $\Delta = \hat{D} + d$ to a pre-existing target-innocent spectrum (KW strata $\hat{D} = \{5, 2, 0\}$, F446), giving $\Delta = \{9, 6, 4\} \rightarrow$ Casimirs $\{45, 12, 0\}$. Given the arena $d=4$ is *forced by the descent*, that map has zero free parameters and hits two targets + one unforced bonus (0-stratum → massless). Not form-cheap. The only residual freedom is the *form* of the map (affine, $+d$) — which the discrete series must force. Still not banked (map constructed, not derived). Count 8.

THE three live count-movers — converge, derive forward

MOVER 1 (nearest) — down-ladder $8 \rightarrow 10$: derive $\Delta = \hat{D} + d$ from the $SO(4,2)$ discrete series · GRACE lead, LYRA + ELIE

Does the $SO(4,2)$ discrete-series lowest-weight at each KW stratum give $\Delta = \hat{D} + d$ *forward* — from the rep theory (or the AdS_5/CFT_4 dimension relation), not because it hits $\{12, 45\}$? Fix the map from the discrete series FIRST, then read the Casimirs. Also close the deposit: the observed ratios $\{20, 45, 900\}$ carry the substrate deposit $d(\nu)=5/3$ (an $SO(5,2)$ quantity) — show it combines with the Casimirs correctly. Bank condition: $\Delta = \hat{D} + d$ forward and deposit combination clean → down-ladder banks, $8 \rightarrow 10$.

MOVER 2 (parallel) — Cabibbo $5/2$: potential mixing bank · LYRA + GRACE

The $5/2 = n_C/\text{rank}$ is target-innocent (fell out of diagonalizing the D_{IV}^5 curvature). Test the holographic dual: does the $SO(4,2)$ 3-point OPE coefficient for the generation strata give $5/2$ forward, matching the bulk curvature ratio? If yes → the mixing $5/2$ banks as a genuine result on the forced arena (bulk-dual). Resolves the mixing arena at the same time.

MOVER 3 (Casey's call) — α $8 \rightarrow 9$

Arena-clarified (Keeper): the Wyler value stays $d=5$ (substrate self-selecting its dimension — no arena error, *supports* K659); the coupling is $d=4$ boundary. So the sweep does not undercut the value-derivation. Reserved to Casey — banks $8 \rightarrow 9$ if he rules the dim-5 value-derivation sufficient.

Cosmology thread (harder, not a near-term count) — Λ interface action

Λ = inter-universe coupling; forward test: $SO(4,2) \rightarrow D_{IV}^5 \rightarrow SO(4,2)$ action = $2^{\{N_c\}} \cdot n_C \cdot g$ factoring (color Fock) $\times$ (bulk) $\times$ (signature). Vision-adjacent; pursue but don't expect a near-term bank.

Per-CI

- GRACE: MOVER 1 — the discrete-series derivation of $\Delta = \hat{D} + d$ (cite the rep theory, don't relabel). Then MOVER 2 (the OPE $5/2$) if cycles remain.

- LYRA: back MOVER 1's theorem layer (is the conformal Casimir the *forced* mass operator? is $\Delta=\hat{D}+d$ the twist/shortening relation of the discrete series?); lead MOVER 2's OPE derivation. Candidate: the arena-principle + operator-pinning may itself be a third structural bank once you show the conformal Casimir is forced (not just fits) — flag it.
- ELIE: ζ / target-innocence check on MOVER 1 (Δ 's forced, not fitted; Five-Absence on the deposit combination) — fires the instant Grace's discrete-series Δ 's land; and the 5/2 target-innocence for MOVER 2.
- KEEPER: adjudicate each mover forward-vs-fit at landing; hold the map-derivation to the discrete series (MOVER 1) and the OPE to forward (MOVER 2); guard peak-convergence (the down-ladder is prettiest now — the bar fires hardest); update the ledger/diagram + the count as movers land.

Reserved for Casey

- α tier-call (K659) — arena supports the value staying $d=5$; $8 \rightarrow 9$ on your ruling.
- Mass-frame ratification — clears the deferred data-layer sync (and will be needed if MOVER 1 banks).

Disciplines armed — the honest way to move the count

We increase the count by DERIVING forward, never by lowering the bar. The down-ladder map is strongly constrained (zero free params given forced $d=4$) but still constructed — banking it before $\Delta=\hat{D}+d$ derives would be a peak-convergence illusion, and a banked-then-retracted count is worse than an honest 8. Δ 's from the discrete series, not picked. One operator per quantity. Cite-don't-relabel (rep theory to the source). Investigate-don't-gate; use linear algebra (it's carried every unlock this arc). Count 8 until a number lands forward with its arena and its map forced.

EOD target

$\Delta = \hat{D} + d$ derived forward from the $SO(4,2)$ discrete series (down-ladder banks $8 \rightarrow 10$) or an honest not-yet with the rep-theory gap named. And/or the OPE 5/2 lands (mixing banks). Count 8 until a forward landing.

— Keeper, 2026-07-06 converge-to-bank. Arena redirect resolved the arc's tangle (object pinned = $SO(4,2)$ conformal Casimir; genus/operator/arena all closed). Down-ladder strongest ever: one zero-free-parameter map $\Delta=\hat{D}+d$ on target-innocent KW strata $\{5,2,0\} \rightarrow \{45,12,0\}$ (2 targets + massless bonus) — not banked, banks when the discrete series forces the map. Three movers: down-ladder ($\Delta=\hat{D}+d$, nearest), Cabibbo 5/2 (OPE dual), α (Casey). Derive forward; count 8 until then.